

TOI-4087b

R-2111

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TOI4087_250319 · created 2026-08-11T00:20:55+00:00

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2460753.8716 +/- 0.0087 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.112 +/- 0.011
Transit depth (Rp/Rs)^2	1.26 +/- 0.25 [%]
Orbital Inclination (inc)	89.72 +/- 1.54
Airmass coefficient 1 (a1)	3.101 +/- 0.026
Airmass coefficient 2 (a2)	-0.7 +/- 0.42
Scatter in the residuals of the lightcurve fit is	0.84 %
Best Comparison Star	#3 - [512, 365]
Optimal Aperture	3.77
Optimal Annulus	10.82
Transit Duration (day)	0.0645 +/- 0.0093

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.112 $\pm$ 0.011 vs 0.10754068997360952 (deviation 0.41 σ)
Duration fit vs published	0.0645 d vs 0.1242 d (deviation 6.42 σ)
Mid-time O-C	2.8 min
Depth SNR	10.2
Residual scatter	0.84 %

Light curve

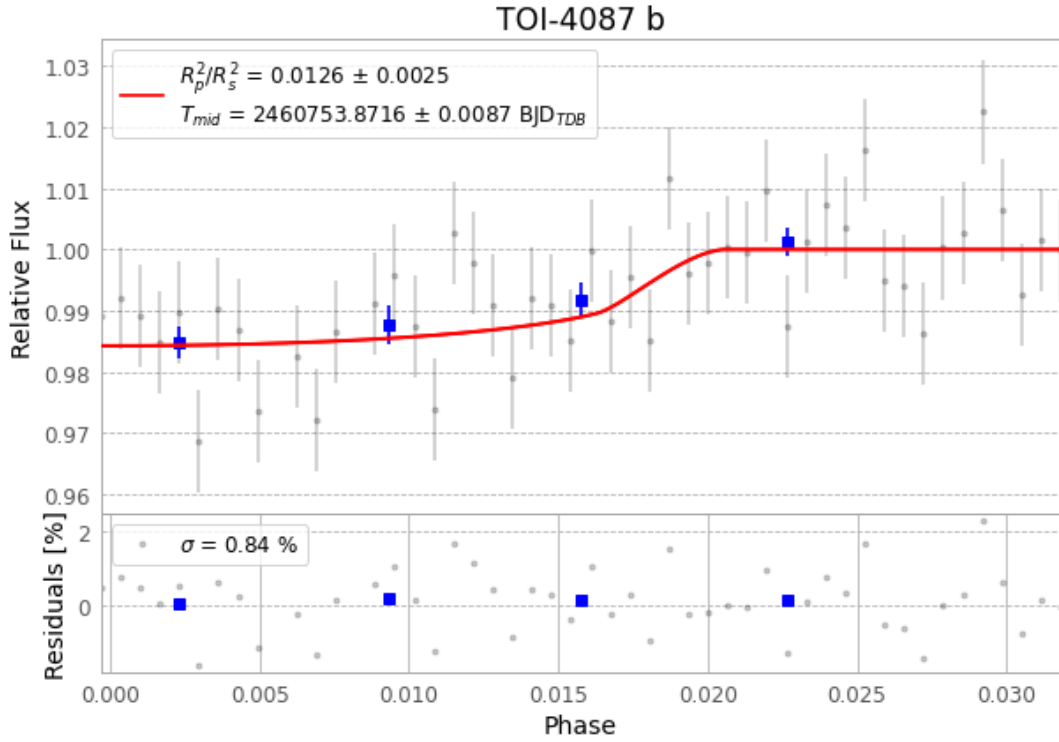


Figure 1 — FinalLightCurve_TOI-4087 b_2025-03-19.png (SHA-256 e051b8adcc2dfc45...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [443, 264], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 e92c69da91908cec...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit fd74a91

Data provenance

50 FITS frames (33 MB), TOI4087250319085115.FITS → TOI4087250319111817.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TOI4087-250319.log (1fdb37820f6e...), FinalLightCurve_TOI-4087 b_2025-03-19.png (e051b8adcc2d...), FinalLightCurve_TOI-4087 b_2025-03-19.csv (7f5082761c16...)

Compute resources

Wall time 126.9 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-11 00:20Z.